O*NET OnLine

Software Quality Assurance Analysts and Testers

15-1253.00

Bright Outlook | Updated 2026

Develop and execute software tests to identify software problems and their causes. Test system modifications to prepare for implementation. Document software and application defects using a bug tracking system and report defects to software or web developers. Create and maintain databases of known defects. May participate in software design reviews to provide input on functional requirements, operational characteristics, product designs, and schedules.

Sample of reported job titles: Automation Tester, Information Technology Analyst (IT Analyst), Quality Assurance Analyst (QA Analyst), Quality Assurance Engineer (QA Engineer), Quality Engineer, Software Quality Assurance Analyst (SQA Analyst), Software Quality Assurance Engineer (SQA Engineer), Software Quality Engineer, Software Test Engineer, Test Engineer

Occupation-Specific Information

Tasks

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- Identify, analyze, and document problems with program function, output, online screen, or content.
- Document software defects, using a bug tracking system, and report defects to software developers.
- Develop testing programs that address areas such as database impacts, software scenarios, regression testing, negative testing, error or bug retests, or usability.
- Design test plans, scenarios, scripts, or procedures.
- Document test procedures to ensure replicability and compliance with standards.

Occupational Requirements

Work Activities

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- **Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.

- **Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
- **Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
- **Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
- **Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.

Detailed Work Activities

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- Document operational activities.
- Analyze data to identify or resolve operational problems.
- Troubleshoot issues with computer applications or systems.
- Compile technical information or documentation.
- Report maintenance or equipment problems to appropriate personnel.

Work Context

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- **E-Mail** — 100% responded “Every day.”
- **Face-to-Face Discussions with Individuals and Within Teams** — 75% responded “Every day.”
- **Spend Time Sitting** — 71% responded “Continually or almost continually.”
- **Importance of Being Exact or Accurate** — 50% responded “Extremely important.”
- **Work With or Contribute to a Work Group or Team** — 58% responded “Extremely important.”

Experience Requirements

Job Zone

Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database

administrators, graphic designers, conservation scientists, art directors, and cost estimators.

SVP Range (7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

State licenses Select a State Go

Apprenticeship Opportunities

All 1 displayed

Example apprenticeship titles for this occupation:

- Software Quality Assurance Tester (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Data base user interface and query software** — Airtable; Apache Hive; Blackboard software; IBM DB2
- **Development environment software** — Apache Kafka; Apache Maven; Apache Subversion SVN; Oracle Java 2 Platform Enterprise Edition J2EE
- **Object or component oriented development software** — Apache Spark; jQuery; Scala; Swift
- **Program testing software** — Hewlett Packard LoadRunner; IBM Rational Robot; JUnit; Selenium
- **Web platform development software** — Django; Google Angular; React; Spring Framework

Hot Technologies are requirements most frequently included across all employer job postings.
See all 126 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 24 In Demand skills for this occupation.

Essential Skills

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- **Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
- **Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
- **Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
- **Speaking** — Talking to others to convey information effectively.
- **Writing** — Communicating effectively in writing as appropriate for the needs of the audience.

Transferable Skills

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- **Programming** — Writing computer programs for various purposes.
- **Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
- **Quality Control Analysis** — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
- **Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
- **Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.

Knowledge

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- **Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
- **English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
- **Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
- **Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.

- **Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

50%	Bachelor's degree required
26%	Associate's degree or other 2-year degree required
9%	Post-secondary certificate required

Worker Characteristics

Abilities

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- **Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
- **Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
- **Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
- **Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
- **Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.

Interests

All 3 displayed

Interest code: **ICR**

Want to discover your interests? Take the **O*NET Interest Profiler**.

- **Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
  - **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
- **Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.

- **Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
- **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.

- **Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
  - **Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.

Work Styles

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- **Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
- **Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
- **Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
- **Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025)	$50.14 hourly, $104,300 annual
State wages	Select a State Go
Local wages	ZIP Code Go
Employment (2025)	187,600 employees
Projected growth (2025-2035)	Faster than average (5% to 6%)
Projected job openings (2025-2035)	10,700
State trends	Select a State Go
Top industries (2025)	Professional, Scientific, and Technical Services Information

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1211.00 Computer Systems Analysts
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1299.04 Penetration Testers
- 15-1252.00 Software Developers
- 17-2112.02 Validation Engineers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- American Society for Quality
- Association for Computing Machinery
- Association of Software Testing
- Computing Research Association
- IEEE Computer Society

Accreditation, Certification, & Unions

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- American Software Testing Qualifications Board ALLY
- Association for Testing and Software Quality Assurance ALLY
- CompTIA
- Institute for Certification of Computing Professionals
- International Software Testing Qualifications Board